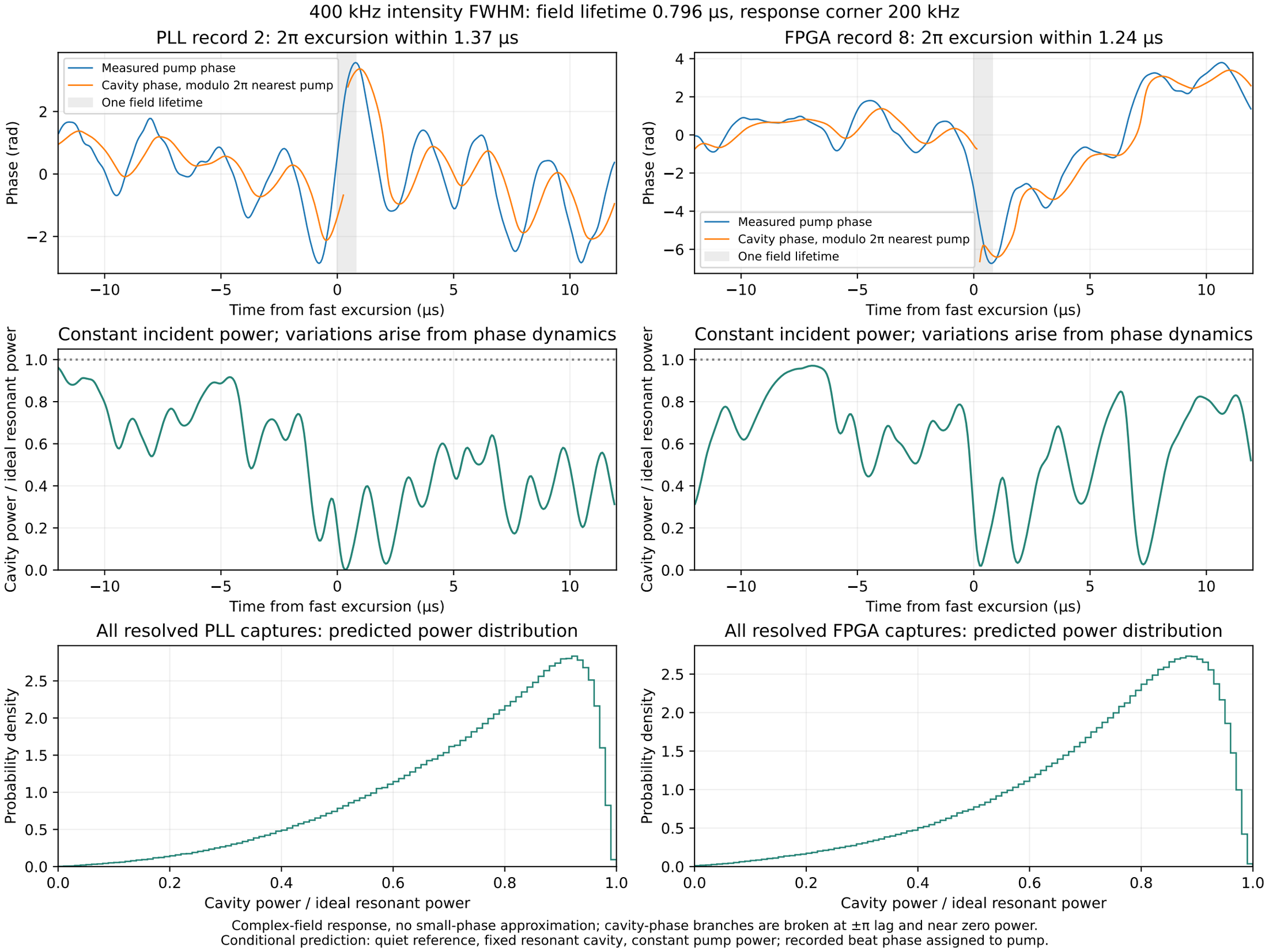
400 kHz intensity FWHM: field lifetime 0.796 µs, response corner 200 kHz
PLL record 2: 2π excursion within 1.37 µs
FPGA record 8: 2π excursion within 1.24 µs
Measured pump phase
Cavity phase, modulo 2π nearest pump
One field lifetime
Phase (rad)
Time from fast excursion (µs)
Constant incident power; variations arise from phase dynamics
Cavity power / ideal resonant power
All resolved PLL captures: predicted power distribution
All resolved FPGA captures: predicted power distribution
Probability density
Complex-field response, no small-phase approximation; cavity-phase branches are broken at ±π lag and near zero power.
Conditional prediction: quiet reference, fixed resonant cavity, constant pump power; recorded beat phase assigned to pump.